Describe a wave

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify amplitude, wavelength, frequency and period.

SUCCESS CRITERIA - I can...

1. Measure amplitude from equilibrium.
2. Measure wavelength between matching points.
3. Connect frequency and period.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “amplitude” and “wavelength” to explain your model or result. Sentence starters are optional.

amplitude: maximum displacement from equilibrium for the wave

wavelength: distance between corresponding points on repeated waves

frequency: number of cycles each second

01 Warm-up: bring back an earlier idea

A 3 kg sample with c=500 J/(kg·°C) warms 5°C. **Find** absorbed energy. If it is 5% of source energy, find source energy.

Word help: absorbed = taken into the specified object or system

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

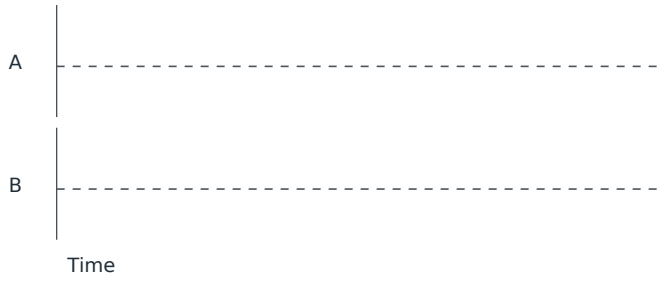
02 Worked example: explain the model

Reference: f = cycles/Δt; T = 1/f.
f is frequency (Hz); T is period (s). λ is crest-to-crest wavelength; amplitude is maximum displacement.

Worked reasoning: A wave makes 6 cycles in 3 s: f=2 Hz and period T=0.5 s. Amplitude measures maximum displacement, not the spacing of crests.

Your turn: Draw one spatial wave and label amplitude and wavelength 0. Separately mark one period on a displacement-time graph.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Describe a wave

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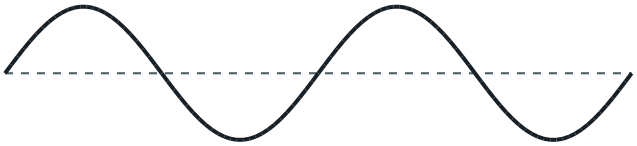
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Label amplitude and one wavelength on the wave diagram.
Explain why crest-to-trough height is not the amplitude.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



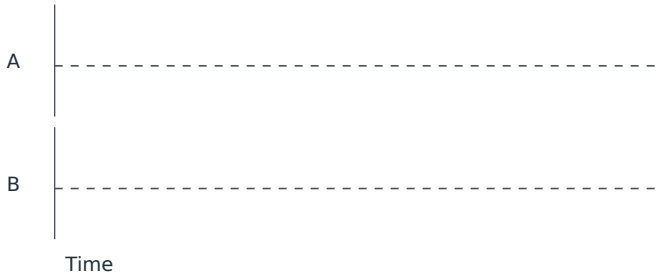
Sketch and label your response.

Marzano check - circle: 1 2 3

04 Apply the physics

A point on a vibrating string completes 15 cycles in 0.60 s. Its crest-to-trough displacement is 0.08 m.
Find frequency, period and amplitude.
Sketch a displacement-time trace and label one period and one amplitude.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

Does increasing amplitude necessarily increase frequency?
Use the definitions to explain 0.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 0.4 kg sample absorbs 2400 J while warming 6°C.
Find specific heat and state a necessary assumption.
Label temperatures, samples and relevant energy transfers.

Word help: specific heat = energy per mass needed for a one-degree temperature change

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

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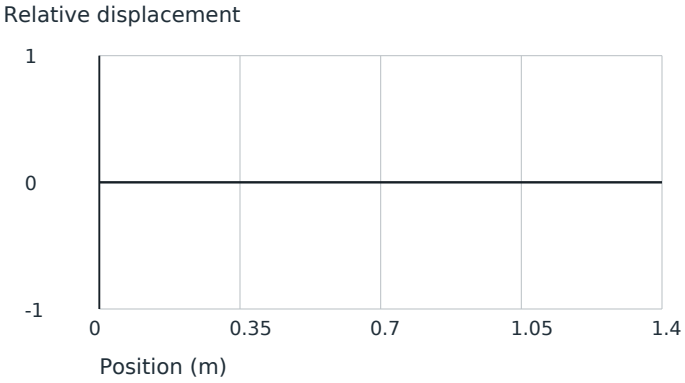
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A sinusoidal snapshot has crests at x=0.20,0.70, 1.20 m.
Find λ. If T=0.10 s, find f.

Sentence starter: I found ... [unit]. This value means ... in this situation.

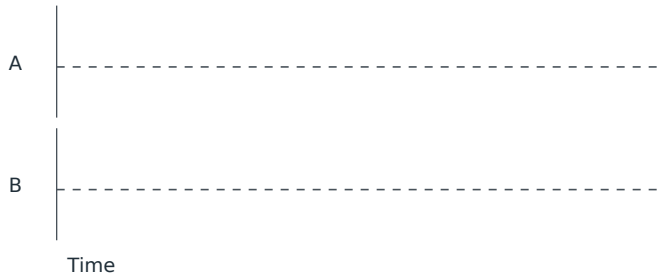


Marzano check - circle: 1 2 3

08 STRETCH: test the model

A point completes 15 cycles in 6 s. Successive crests are 1.2 m apart.
Find period, frequency and wave speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3